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7/12/25

Applied Cyber Threat Analysis – NMAP/Nessus Project

- **This lab was conducted within the Virginia Cyber Range Environment.**
- **Report based off assignment assigned by George Mason University - Department of Information Sciences and Technology College of Engineering and Computing**
- **Cheche Agada. (2024). Mini Project IT-462: Applied Cyber Threat Analysis, George Mason University**

Introduction:
This lab will guide you through the process of installing a hypervisor (VMWare), and two virtual machines – Windows and bee-box.

bee-box is **a custom Linux VM pre-installed with bWAPP (buggy web application)**. It is intentionally pre-installed with a vulnerable application to provide the opportunity to explore application vulnerabilities in a test environment. The VM gives several opportunities to hack and deface the bWAPP website that is contained within it. However, this exercise will only focus on assessing the vulnerabilities, not exploiting them.

You will use Nmap/ZenMap and Nessus to scan applications and systems for specific vulnerabilities like open ports and other software vulnerabilities.

Objective:
Set up a virtual environment on your personal device and conduct vulnerability assessment exercises.

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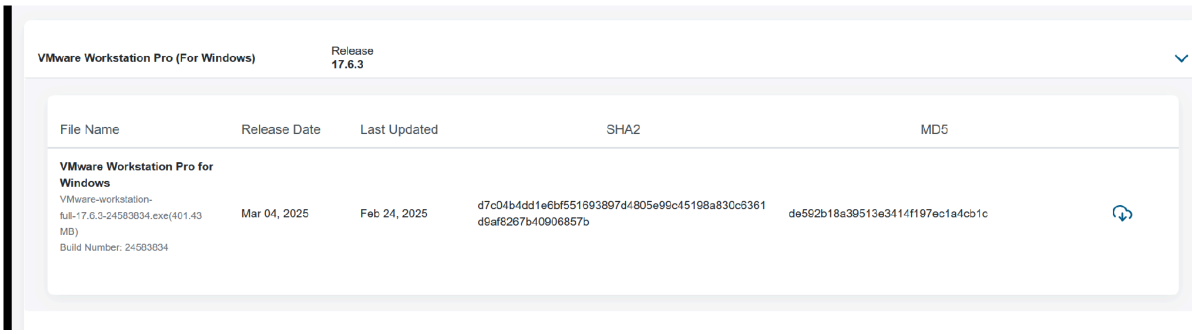
Part 1 - Setup

Task 1 – Install VMWare

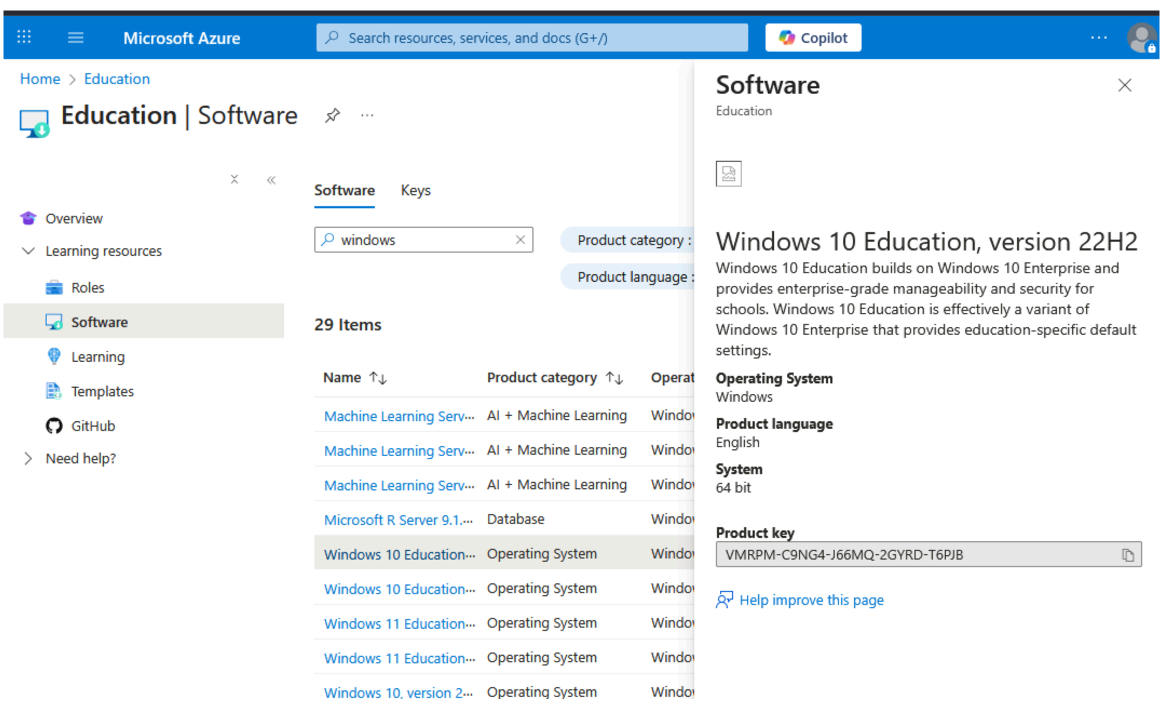
Task 2 – Install a Windows Virtual Machine

Lab Exercise 1

- VMWare version **17.6.3**



Windows 10 Education, version 22H2



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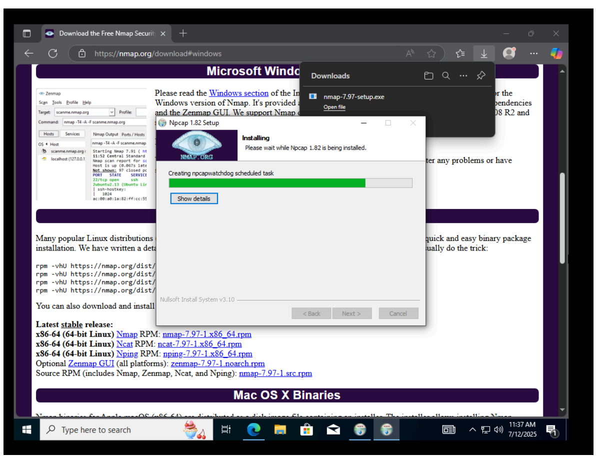
PART 2

Overview

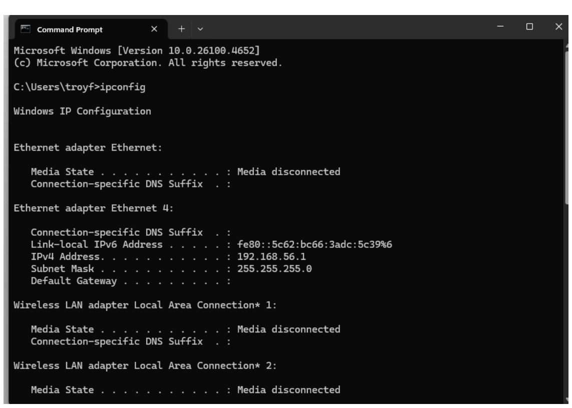
In this section, you will conduct security assessment tasks to check for system and software vulnerabilities. We will use Nmap/ZenMap to check for open ports and services in order to determine vulnerable or exploitable services. Then we will use Nessus to scan a buggy application installed on a virtual machine to check for vulnerabilities in the application. **Remember to include screenshots to show each step.**

Task 3A: Conduct an Nmap Scan of Your Host Computer using ZenMap

- Download and install Nmap on your Windows virtual machine (VM) (https://nmap.org/download)
- ZenMap is the GUI version of Nmap – they are installed together.



- On your host computer, open a command prompt and obtain the IP (Internet Protocol) address.



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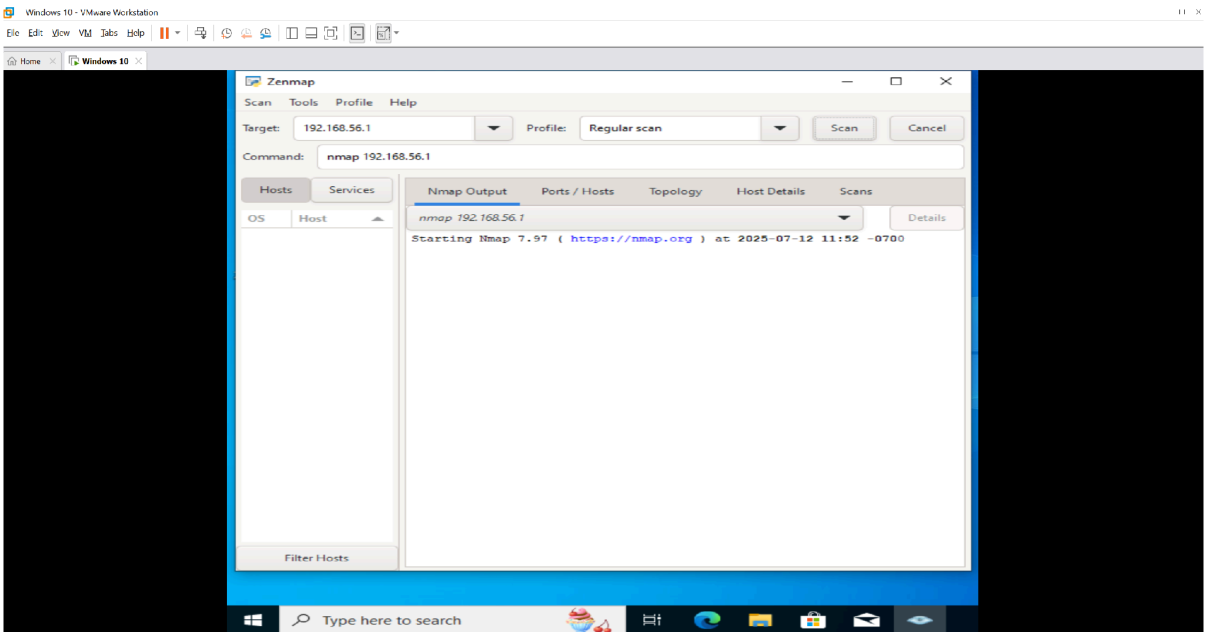
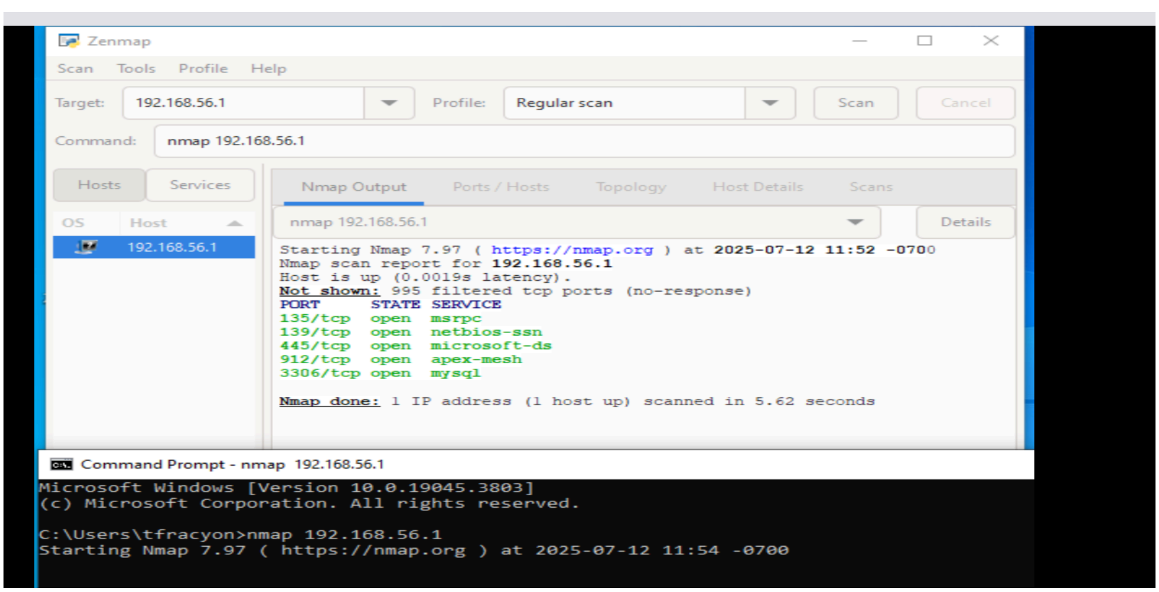
- On your VM, start the Nmap program.
- Provide the IP address of your host machine in the space provided (**Target**).
- In the **Profile** section, select "Regular scan" as the type of scan to conduct.
- Nmap generates the required command as shown in the **Command** section.
- Select the **Scan** button to start the scan.



Task 3B: Conduct an Nmap Scan of Your Host Computer (Command Line)

- Run the same command from (3A) on the command prompt of your VM using Nmap.



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Lab Exercise 2

(1) Compare the output of 3A and 3B. Are they the same? Why or why not?
They are the same because they both list the port #, the state of the service, and the name of the service.

(2) How many ports are open? **5 ports**

(3) Select an open port and investigate the service hosted at this port. Is it safe to have the selected port open? What risk is associated with having this port open?
Port 3306 is associated with mysql. It is considered unsafe to have this port left open because of unauthorized personnel getting access.

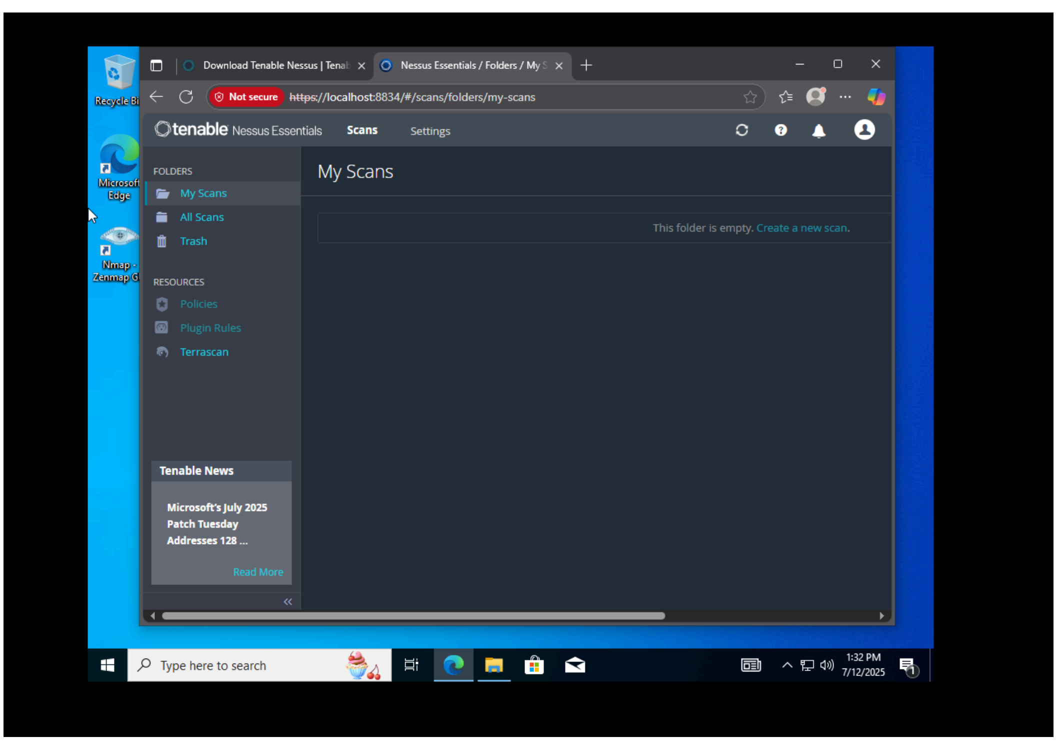
(4) How many ports are filtered? **995 filtered tcp ports.**

(5) What does a filtered port mean? **A firewall is blocking access.**

Task 4: Install Nessus

- Download and install Nessus Essentials on your Windows VM (https://www.tenable.com/products/nessus/nessus-essentials)

- Program will run in your browser (https://localhost:8834)



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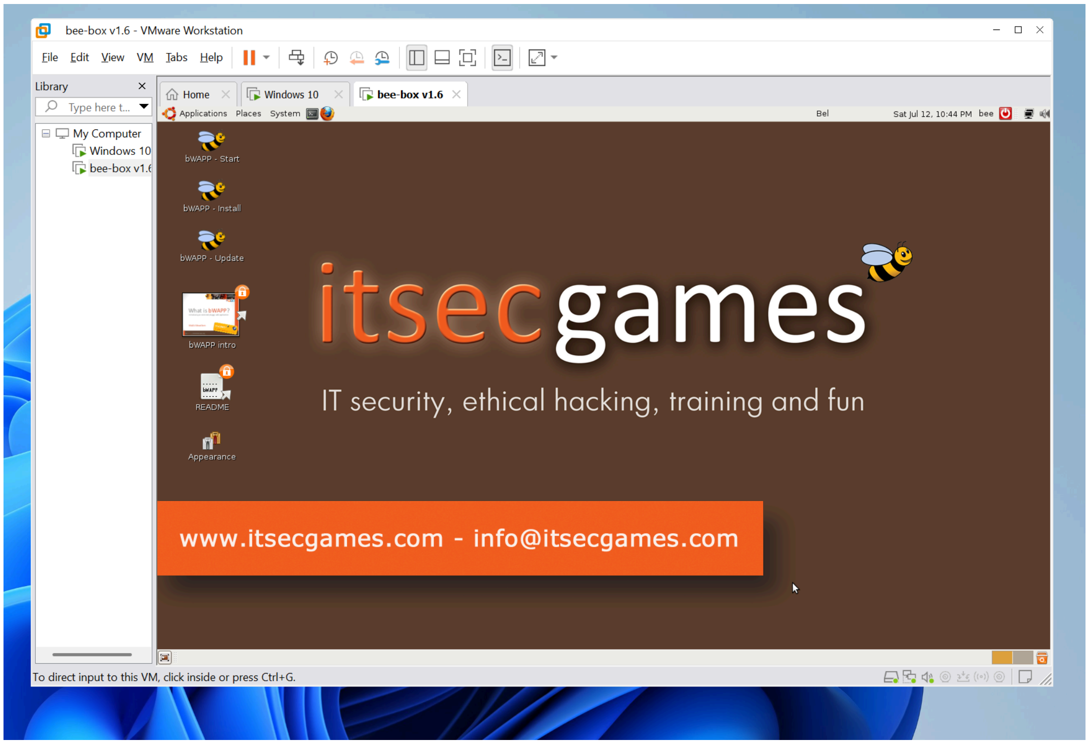
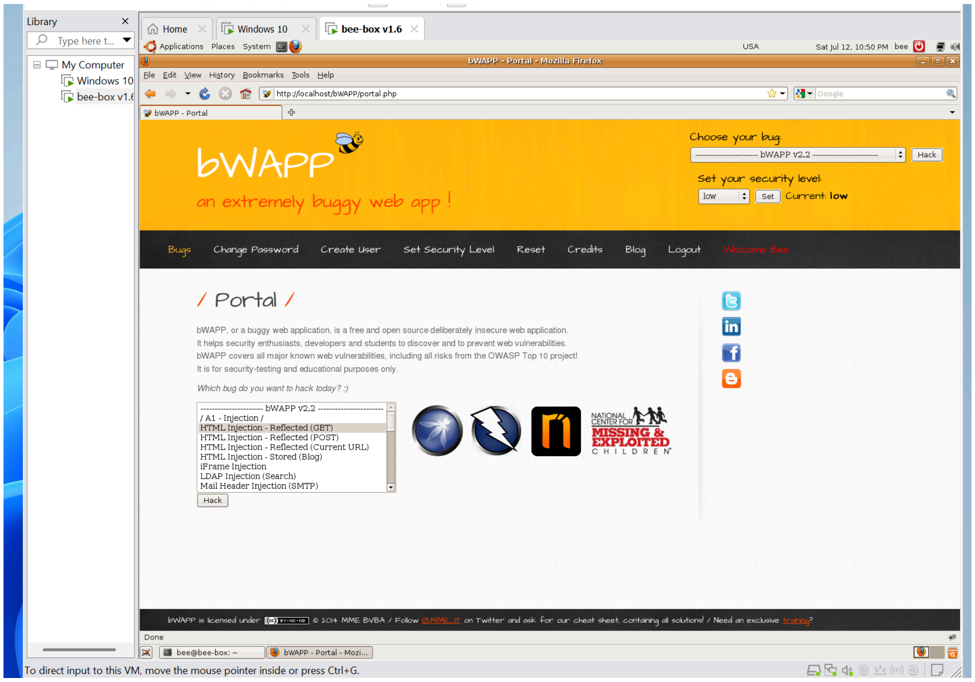
Task 5: Set up bee-box

- Download the buggy Web Application (bWAPP) VM – **bee-box** – **directly to your computer**. You must download it from the **Mini Project** folder on Blackboard.
- Unzip the folder and open the bee-box VM using your hypervisor (VMWare).





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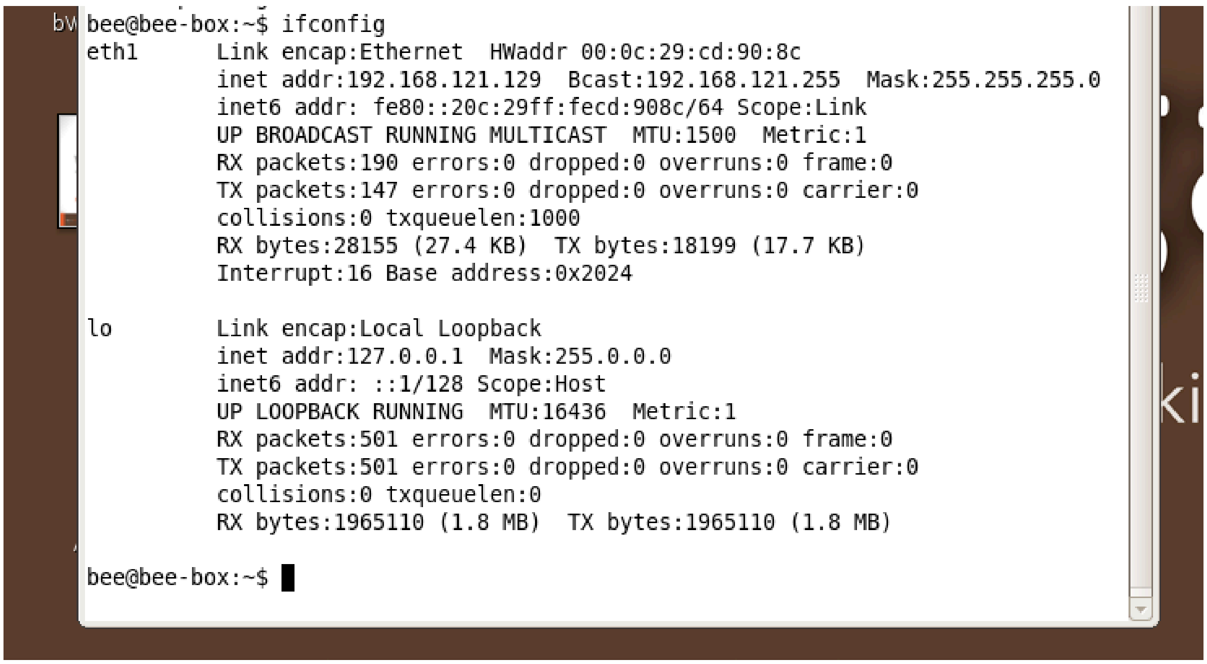
- If requested, provide username (**bee**) and password (**bug**) to login.

- Open a terminal window and change the language preference (i.e. **System –> Preferences –> Keyboard –> Layouts**) to USA. You may have to remove and re-add ‘**USA**’ for your keyboard to work correctly.
- Obtain the IP address of the bee-box using **‘ifconfig’** on the terminal.



Task 6A: Conduct an Unauthenticated Nessus Scan

- Start the Nessus scanner in your browser (https://localhost:8834)
- Select New Scan -> Advanced Scan
- Provide a name for your scan “**Name**”
- Provide the IP address of the VM to scan (i.e. IP address of the bWAPP VM) “**Targets**”
- Click the arrow on the right side of the “**Save**” button and select “**Launch**” to start the scan
- After the scan is complete, examine the report.

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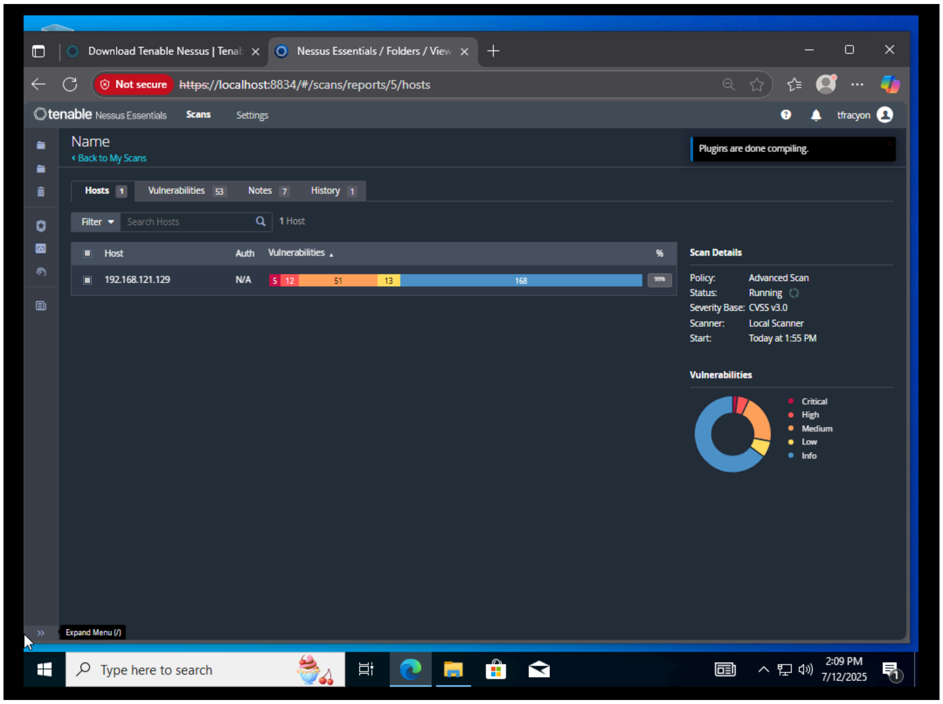

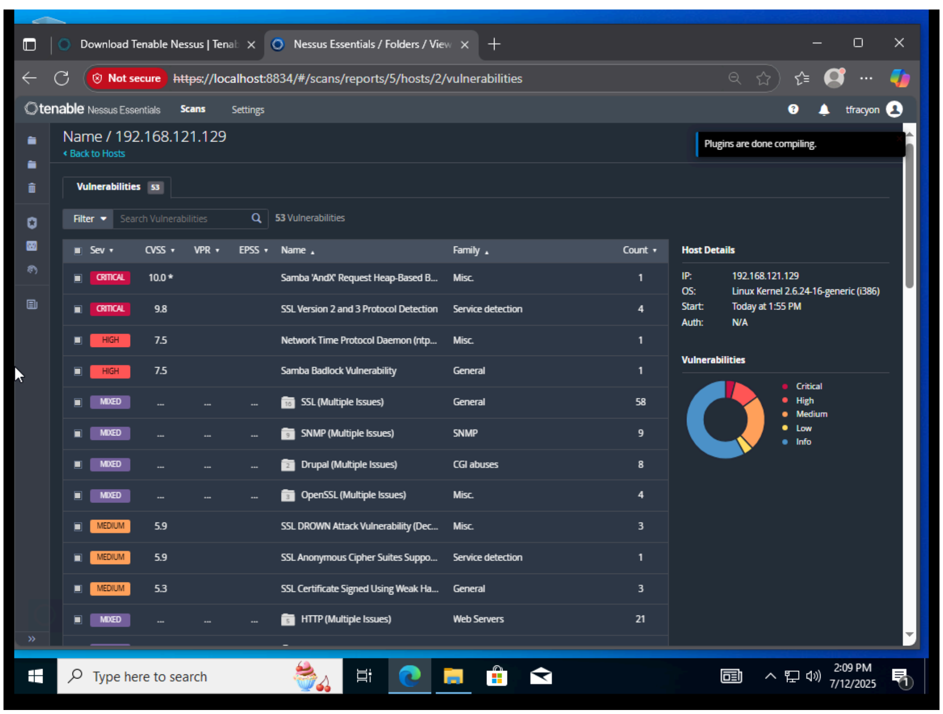

Task 6B: Conduct an Authenticated Nessus Scan

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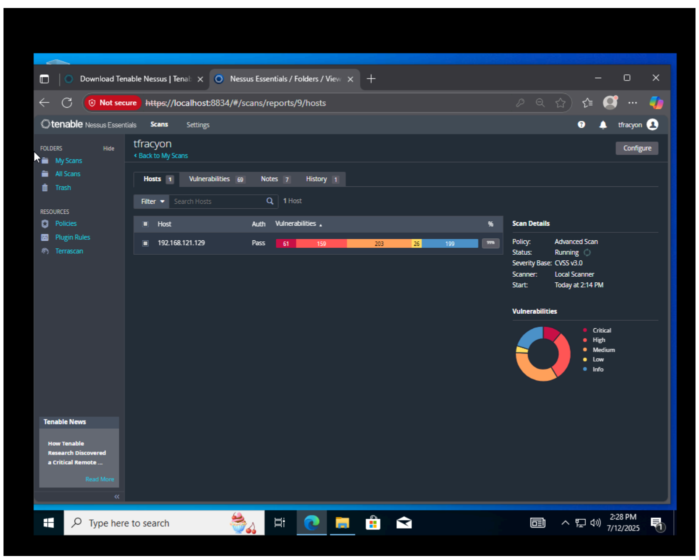
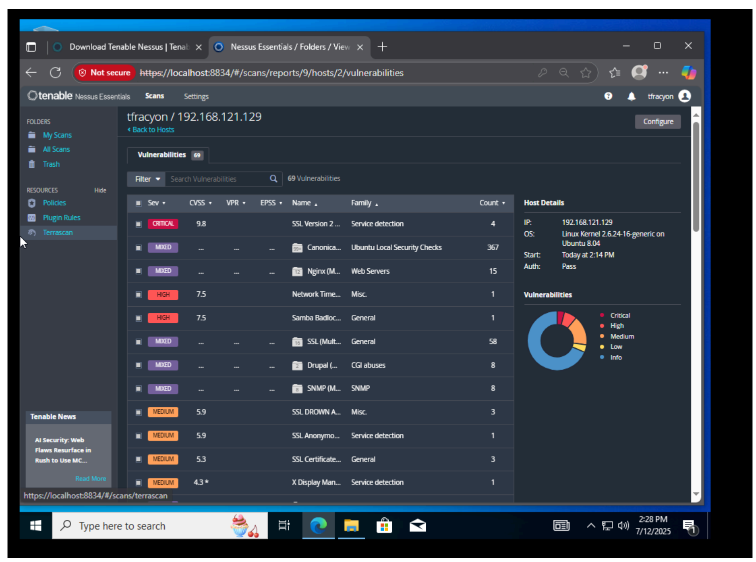
- Start the Nessus scanner in your browser (localhost:8834)
- Select **New Scan** -> **Advanced Scan**
- Provide a name for your scan “**Name**”
- Provide the IP address of the VM to scan (i.e. IP address of the bWAPP VM) “**Targets**”
- Click on the “Credentials” tab and select “**SSH**”
- Select “**password**” from the drop-down menu beside “**Authentication method**”
- Provide the username (**bee**) and password (**bug**)
- Select “**sudo**” from the drop down beside “**Elevate privileges with**”
- Click the arrow on the right side of the “**Save**” button and select “**Launch**” to start the scan
- After the scan is complete, examine the report.





Lab Exercise 3

(1) Compare the output of Task 6A and Task 6B. Are they the same? Why or why not?
They are not the same, there is a significant increase in the numbers of all types of vulnerabilities. There is a larger ratio of the more severe vulnerabilities. There are also more remediation recommendations on the 6B scan.

(2) Include screenshots for the outputs for Task 6A and Task 6B

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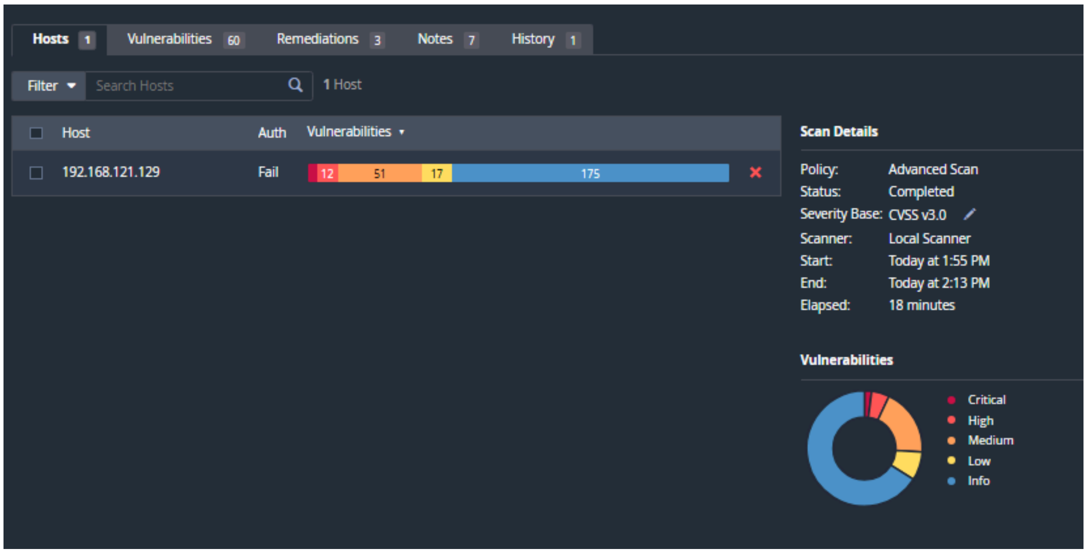

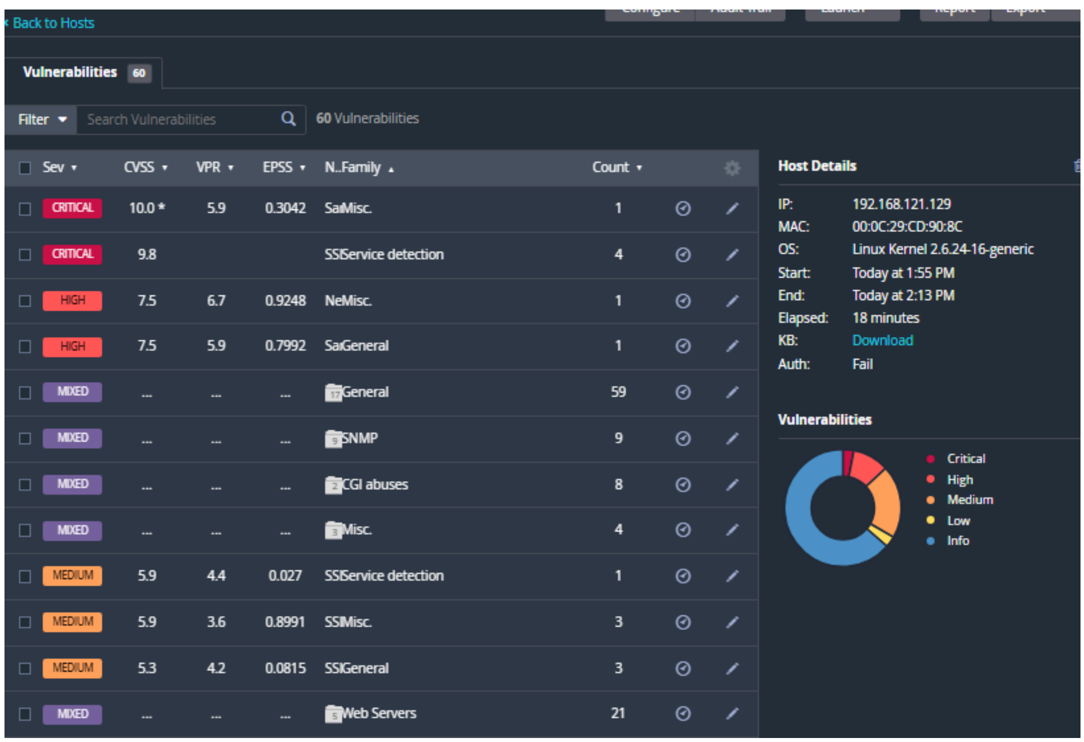

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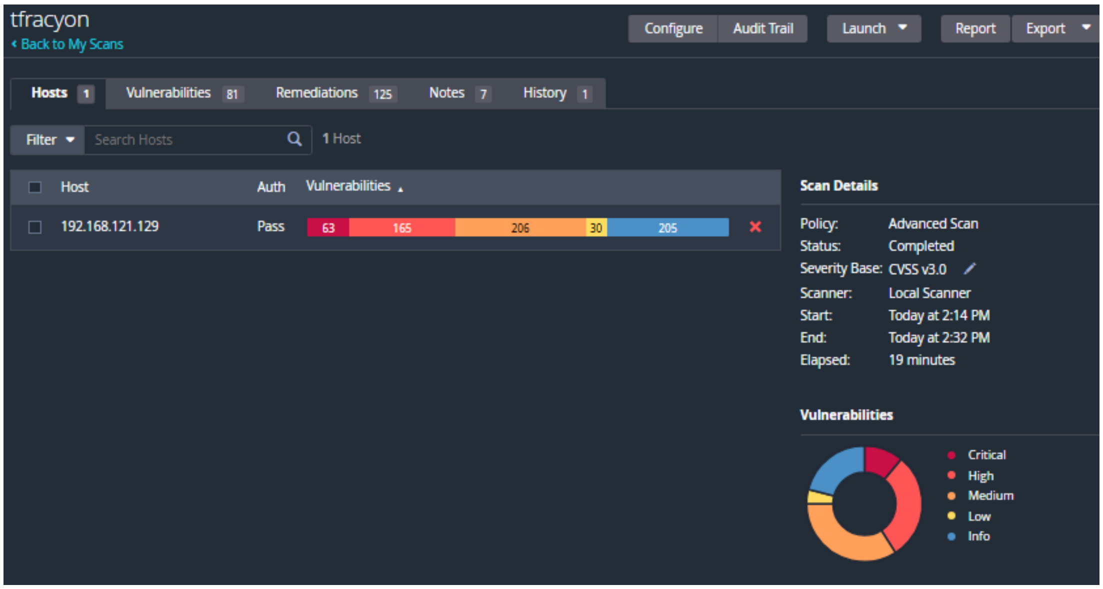

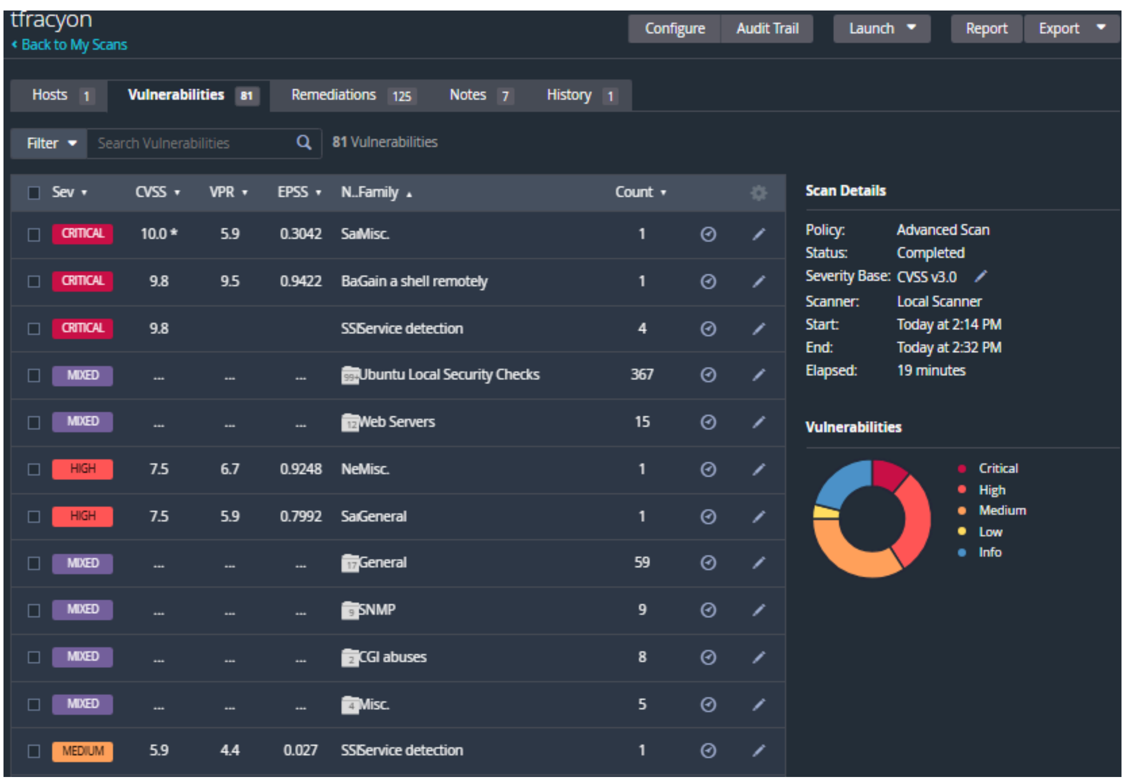

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Resources

- Creating Windows 10 VM in VMWare
  https://www.youtube.com/watch?v=Gf2-p7VzbMg
  https://www.youtube.com/watch?v=O4-9YJD_XNg

- Nessus
  https://www.youtube.com/watch?v=rSGGrbpL7dc
  https://www.youtube.com/watch?v=xJ9k1ksgA9Y
  https://www.youtube.com/watch?v=EIqyY_3f0zk